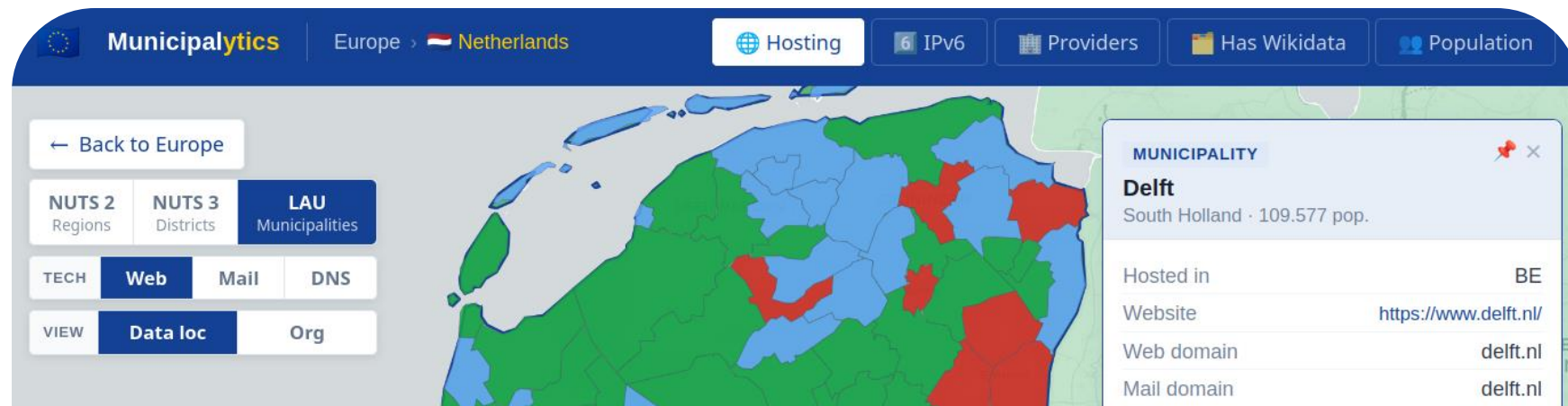




Municipalytics: Mapping the Digital Infrastructure of Municipalities Across GDPR Countries

Florian Holzbauer, Gabriel Gegenhuber and Johanna Ullrich

Digital Sovereignty

WORLD • INTERNATIONAL JUSTICE

Nicolas Guillou, French ICC judge sanctioned by the US: 'You are effectively blacklisted by much of the world's banking system'

Le Monde

FRIDAY, 22 MAY 2026 - 11:10

Microsoft accused of leaking Dutch civil servants' names to U.S. government

NL#TIMES

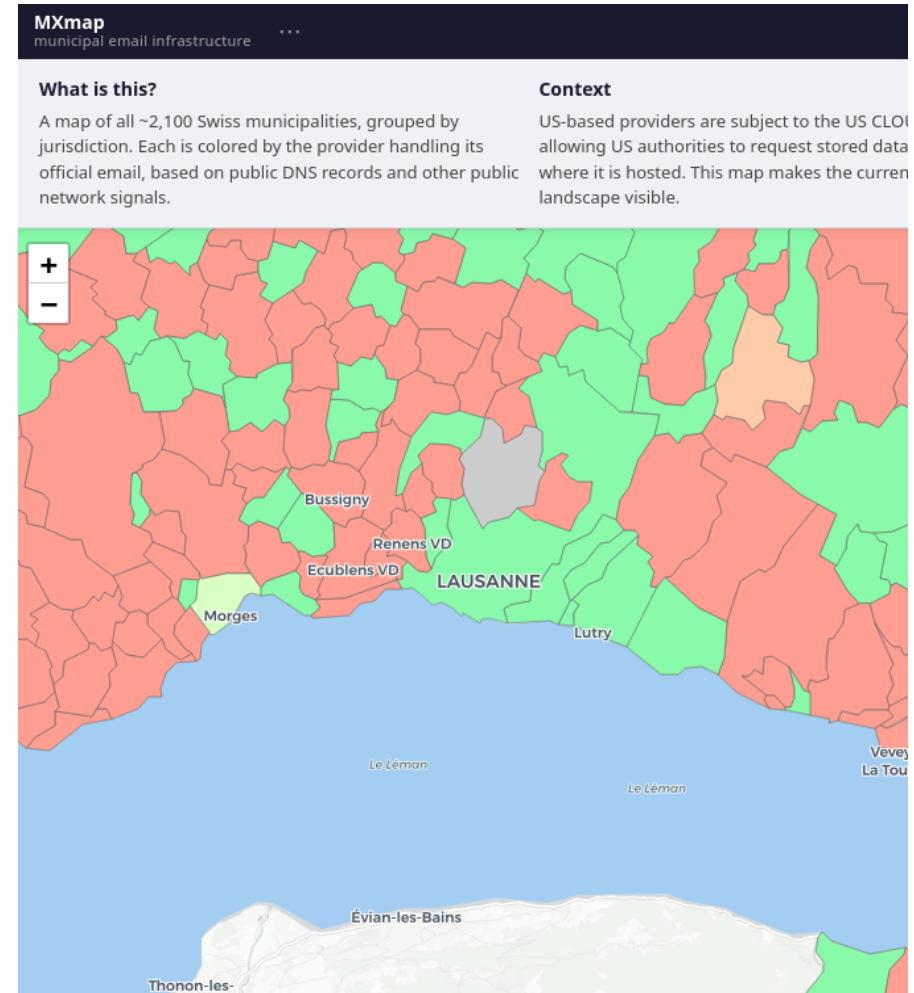
+ email content

Availability

Confidentiality

Introduction

MXMap shows dependency of Swiss municipalities (credits David Huser)



Red = emails end up in US cloud

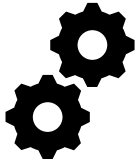
Introduction

RQ: What is the current state of municipal hosting across GDPR countries?

Introduction



Eurostat provides data on the local administrative layer across Europe.

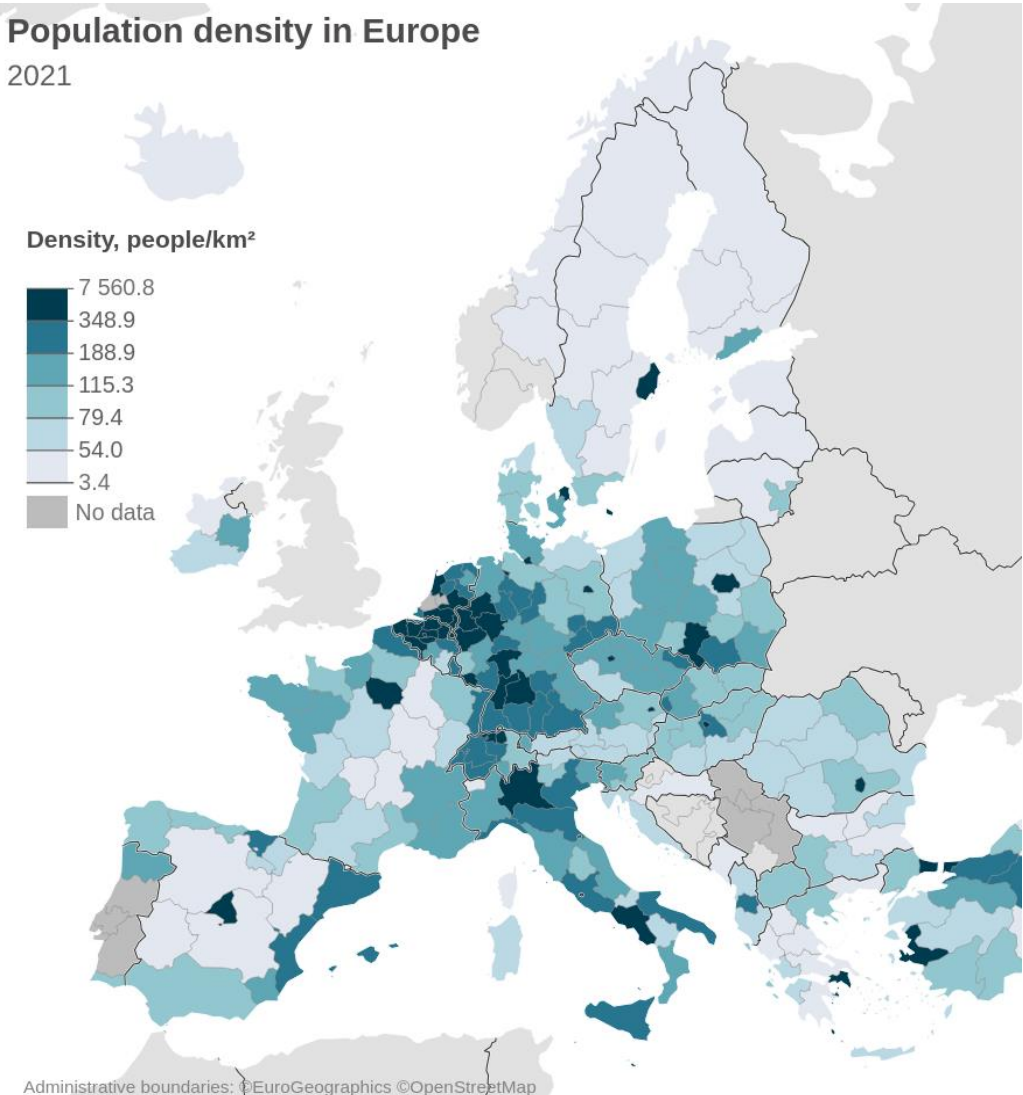
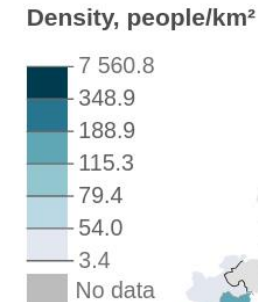


There is no such data collected for **digital infrastructure**. We miss a comprehensive **domain list**.



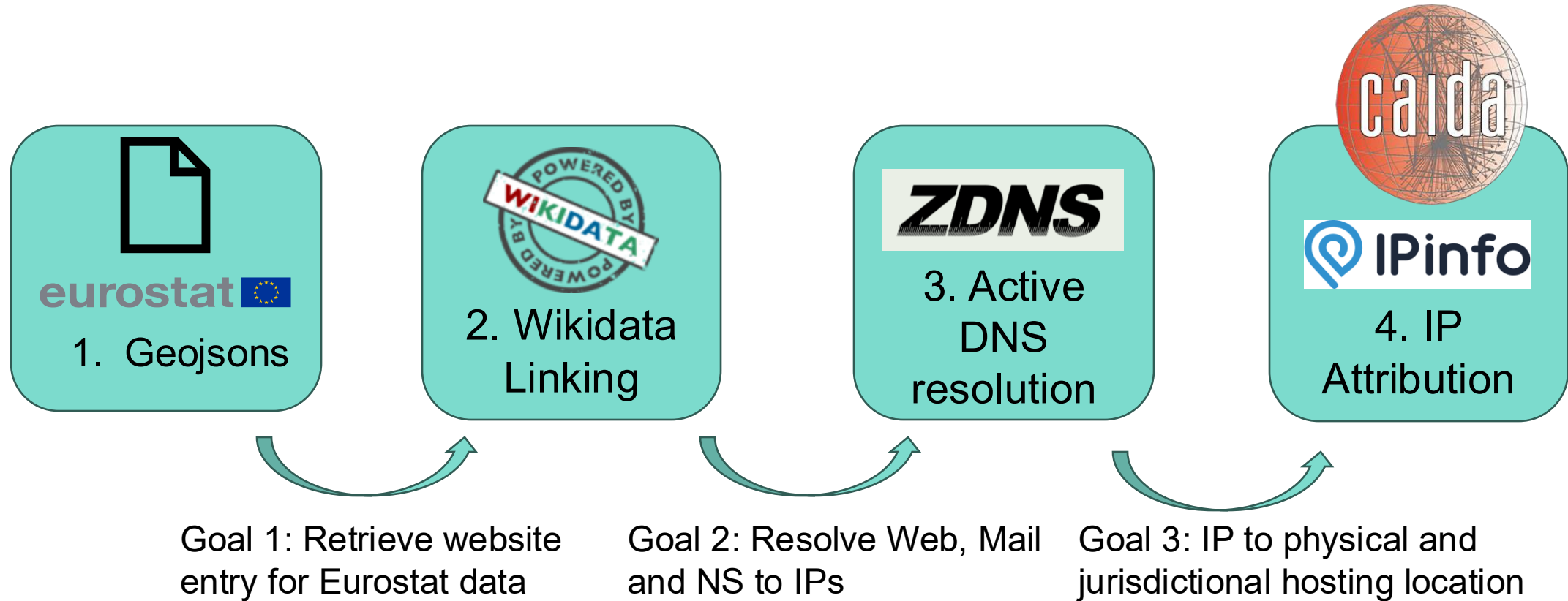
Link Eurostat administrative units with **municipal wikidata** entries.

Population density in Europe
2021



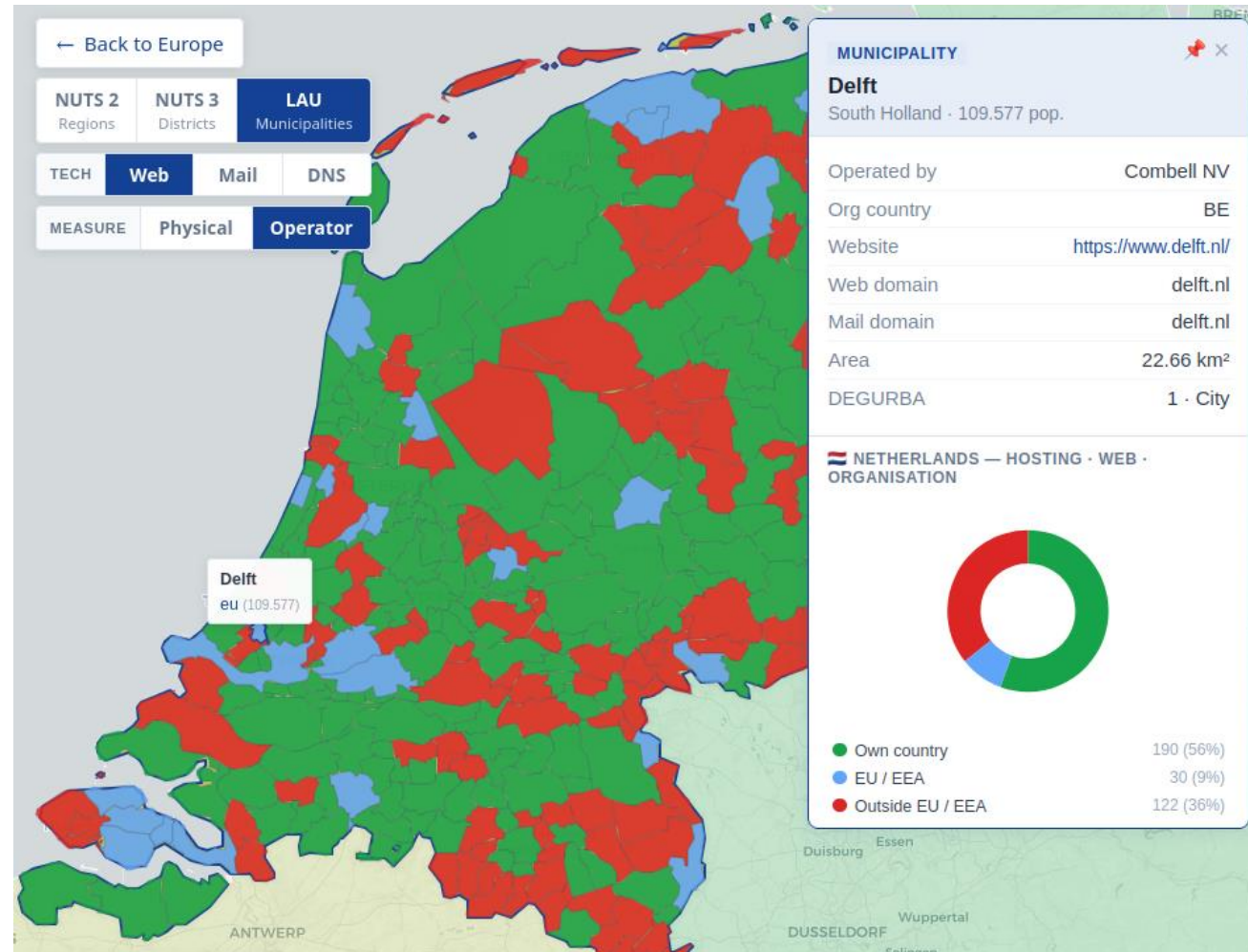
Administrative boundaries: ©EuroGeographics ©OpenStreetMap

Methodology Overview



Data Overview: Delft

Overall Goal:
Reproducible and
automated pipeline for
all countries



Data Overview: Delft

1. - **eurostat** 
- *Two Geojson Inputs:*

- *NUTS regions*

- *LAU's*

	Category	Example
Eurostat	Country	 Netherlands
	NUTS-2 region	Zuid-Holland (NL36)
	NUTS-3 region	Delft en Westland (NL362)
	LAU	Delft
	GISCO ID	NL_GM0503
	Population	109,577
	Population density	4556.2 / km ²
	DEGURBA	1 - City
	Area	22.66 km ²

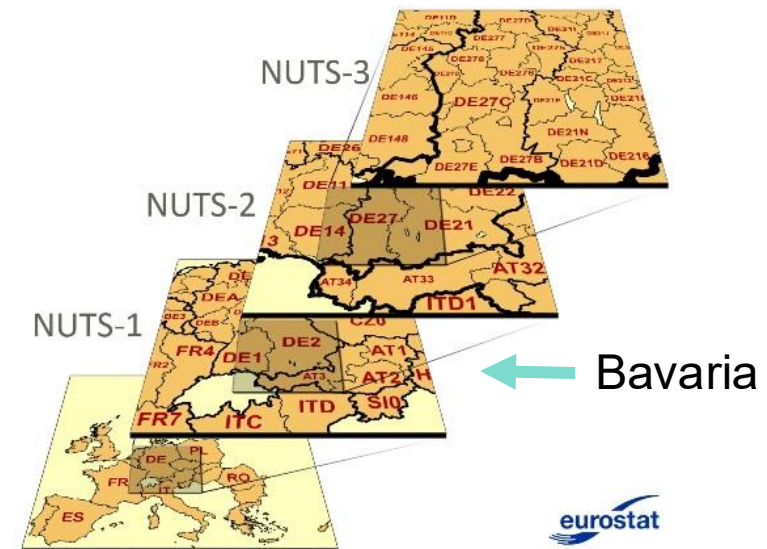
Data Overview: Delft

1. - eurostat □ *NUTS = Nomenclature of territorial units for statistics*

- 4/5 (=LAU's) ... municipality/commune
- 3... district
- 2... province
- 1... state

Category	Example
Country	 Netherlands
NUTS-2 region	Zuid-Holland (NL36)
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LAU	Delft
GISCO ID	NL_GM0503
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Eurostat



Data Overview: Delft

1. - **eurostat** 
 - *NUTS = Nomenclature of territorial units for statistics*

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LAU	Delft
GISCO ID	NL_GM0503
Population	109,577
Population density	4556.2 / km ²
DEGURBA	1 - City
Area	22.66 km ²

Eurostat

```
"type": "Feature",
"properties": {"NUTS_NAME": ".." },
"geometry": { },
  # either 4326 unprojected lon,lat
  # or 3035 projected x,y
```

Data Overview: Delft

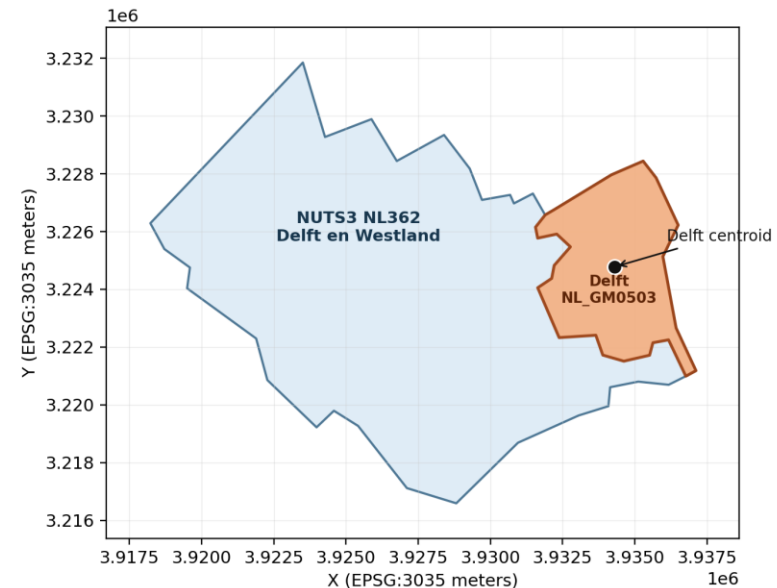
1. - **eurostat** 
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Country	 Netherlands
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Eurostat



Data Overview: Delft

1. - eurostat 

2. Wikidata Linking

- Goal: List of Website/Email entries

	Category	Example
Eurostat	Country	 Netherlands
	NUTS-2 region	Zuid-Holland (NL36)
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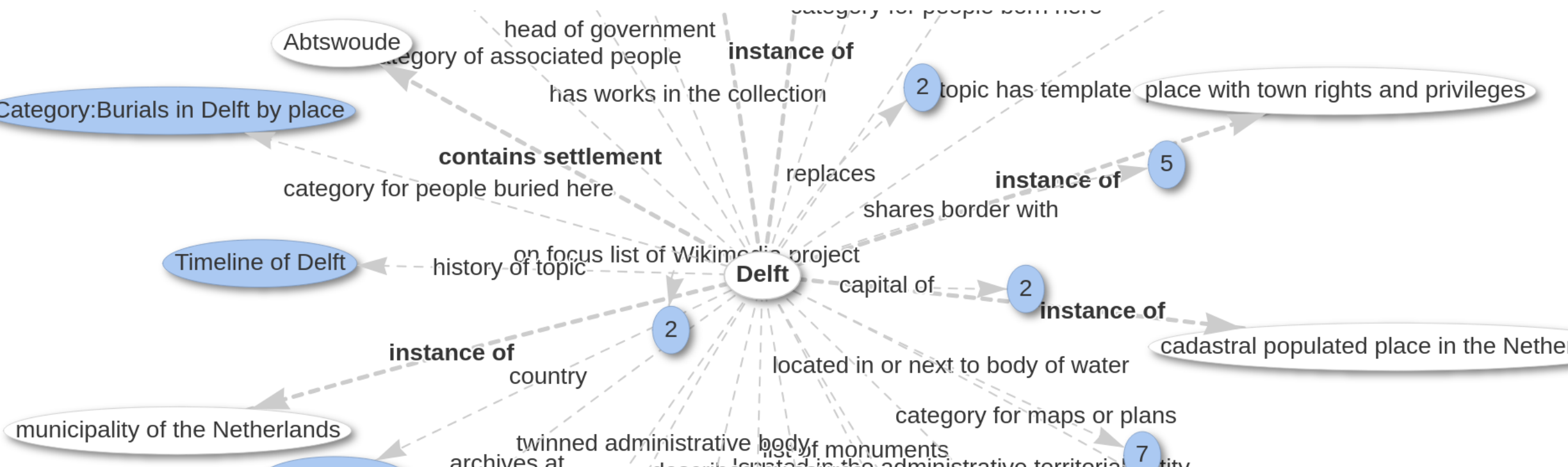
Eurostat 

What domain is Delft using for web/email?

Wikidata Data Model

Subject → predicate → object

Delft → country → Netherlands



Data Overview: Delft

1. - **eurostat** 

2. Wikidata Linking

- Method 1: Property match based on GISCO ID
- Method 2: Spatial Lookup
 - For countries without matching GISCO ID
 - Delft (Q690) > instance of (P31) > municipality of the Netherlands(Q2039348)
 - 1. Retrieve all the entries of a country
 - 2. Match LAU names, use spatial lookup for multiple results based on Wikidata coordinates.

Data Overview: Delft

1. - eurostat

2. Wikidata Linking

- Goal: find Website/Email entries
- Bridge:
P382 ... CBS municipality code



Full property list in paper

	Category	Example
Eurostat	Country	 Netherlands
	NUTS-2 region	Zuid-Holland (NL36)
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	LAU	Delft
	GISCO ID	NL_GM0503
	Population	109,577
	Population density	4556.2 / km ²
Wikidata	DEGURBA	1 - City
	Area	22.66 km ²
	Wikidata property	P382
	Matched code	0503 (GISCO ID derived filter)
	Wikidata QID	Q690
	Website	https://www.delft.nl/

Data Overview: Delft

1. - eurostat

2. Wikidata Linking

- Website/Email entries
- P382 ... CBS municipality code

	Category	Example
Eurostat	Country	 Netherlands
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	Matched code	0503 (GISCO ID derived filter)
	Wikidata QID	Q690
	Website	https://www.delft.nl/

3. Wikidata Query Service (SPARQL)

```
SELECT ?item ?itemLabel ?website ?email WHERE {
  ?item wdt:P382 "0503" .
  OPTIONAL { ?item wdt:P856 ?website . }
  OPTIONAL { ?item wdt:P968 ?email . }
```

Data Overview: Delft

1. - eurostat 
2. Wikidata Properties
 - Website/Email entries
3. DNS resolution

	Category	Example
Eurostat	Country	 Netherlands
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	DEGURBA	1 - City
	Area	22.66 km ²
Wikidata	Wikidata property	P382
	Matched code	0503 (GISCO ID derived filter)
	Wikidata QID	Q690
	Website	https://www.delft.nl/
ZDNS	 A	185.135.13.93
	 AAAA	2a06:efc0:0:103::32
	 MX	m1.delft.nl → 145.32.97.12
	 SPF	v=spf1 ...
	 NS	ns.delft.nl → 145.32.97.4

Data Overview: Delft

1. - **eurostat** 
2. Wikidata Linking
 - Website/Email entries
3. DNS resolution
4. Physical Hosting Loc.
 - IP to country-level geolocation

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Eurostat	Country	 Netherlands
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	 AAAA	2a06:efc0:0:103::32
	 MX	m1.delft.nl → 145.32.97.12
	 SPF	v=spf1 ...
	 NS	ns.delft.nl → 145.32.97.4
IPinfo	 Location	 Belgium
	 Location	 Netherlands
	 Location	 Netherlands

Data Overview: Delft

1. - **eurostat** 
2. Wikidata Properties
 - Website/Email entries
3. DNS resolution
4. Physical Hosting Loc.
5. HQ Jurisdiction

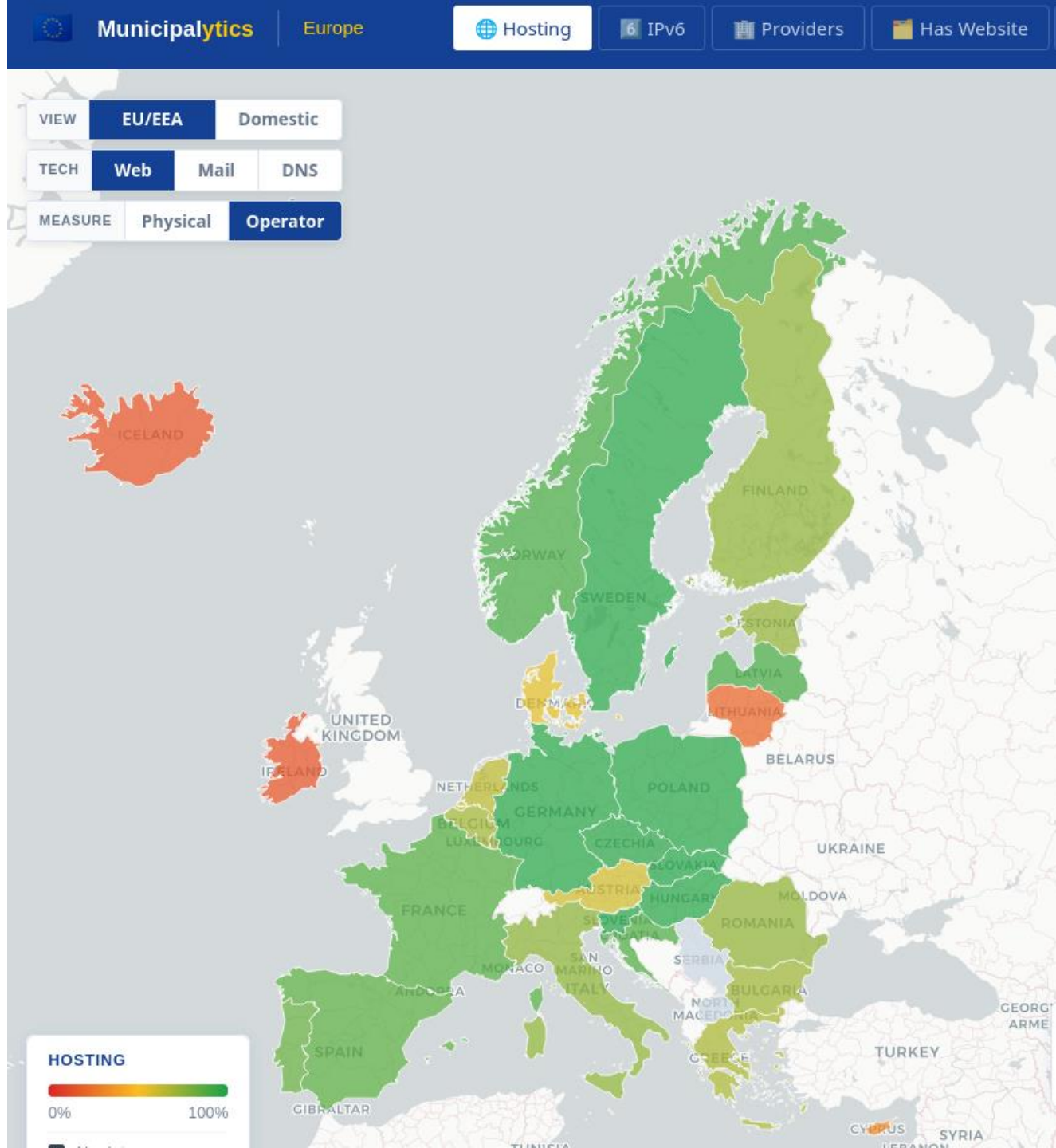
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	 AAAA	2a06:efc0:0:103::32
	 MX	m1.delft.nl → 145.32.97.12
	 SPF	v=spf1 ...
	 NS	ns.delft.nl → 145.32.97.4
IPinfo	 Location	 Belgium
	 Location	 Netherlands
	 Location	 Netherlands
CAIDA AS	 ASN	AS34762
	 Org.	Combell NV
	 Org. Jurisd.	 Belgium
	  ASN	AS34108
	  Org.	Stichting Breedband Delft
	  Org. Jurisd.	 Netherlands

Municipal Coverage

Category	#
Countries	30 (EU27 + EEA)
Eurostat LAU's	95,498
Wikidata coverage	88,948 (93%)
With website entry	69,220 (73%)
% Resolvable	97% NS – 81% SPF

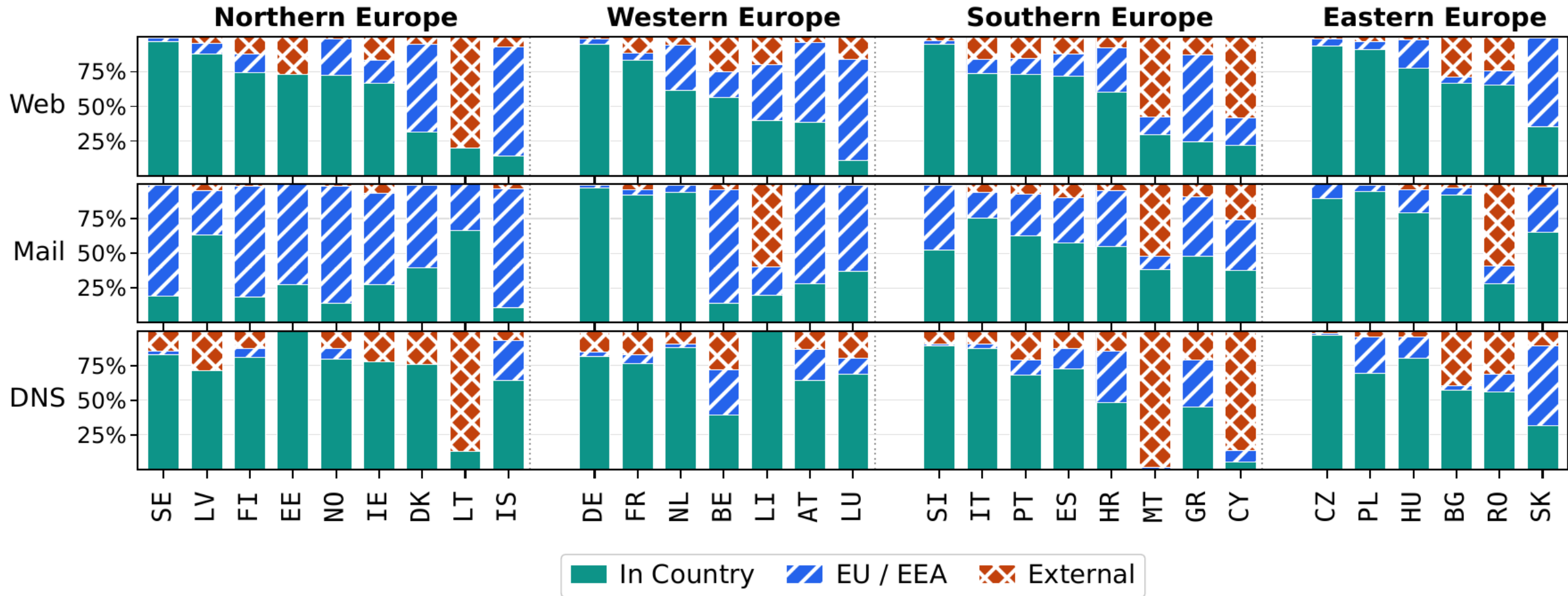
30 Countries: EU 27 + EEA
 GDPR applies also to EEA countries

CC	LAUs
 AT	2,093
 BE	581
 BG	265
 CY+	615
 CZ	6,258
 DE	10,978
 DK	99
 EE	79
 ES	8,132
 FI	309
 FR	34,946
 GR+	6,142
 HR	556
 HU	3,155
 IE+	166
 IS	64
 IT	7,900
 LI	11
 LT	60
 LU	100
 LV	43
 MT	68
 NL	342
 NO	357
 PL	2,477
 PT+	3,092
 RO	3,181
 SE	290
 SI	212
 SK	2,927
TOTAL	95,498

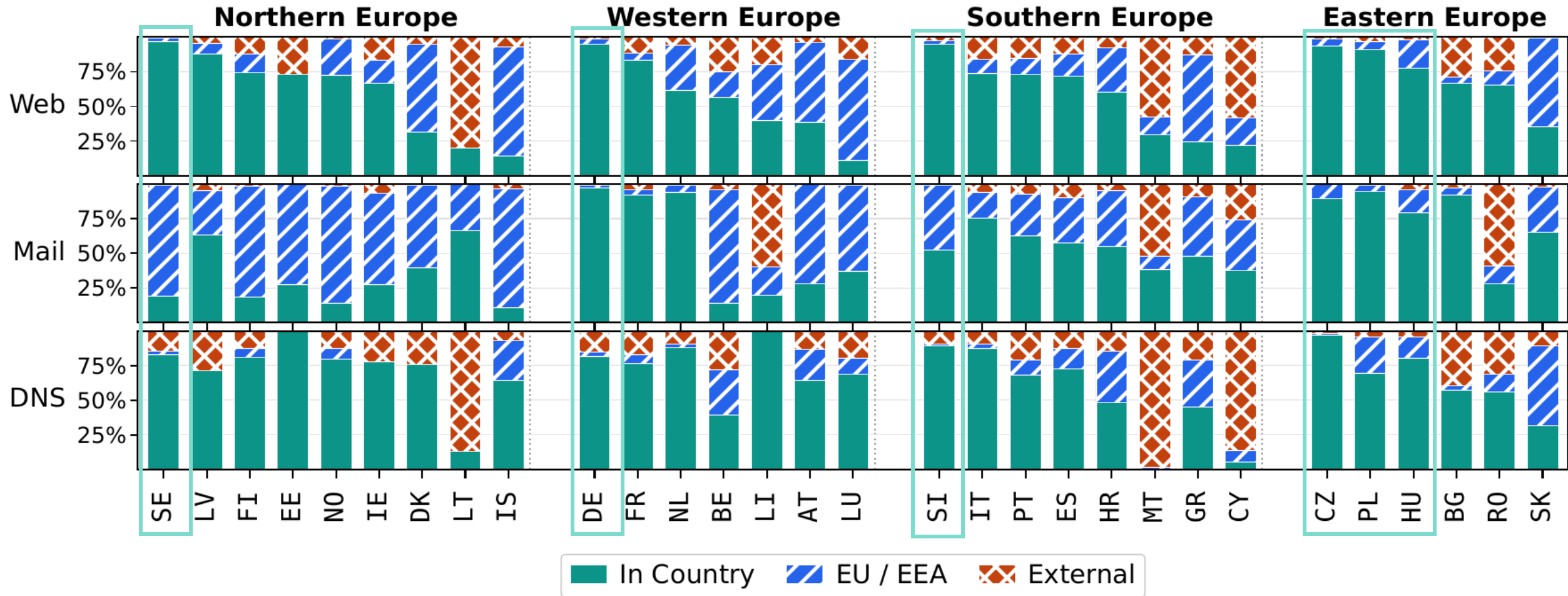


Evaluation and Results

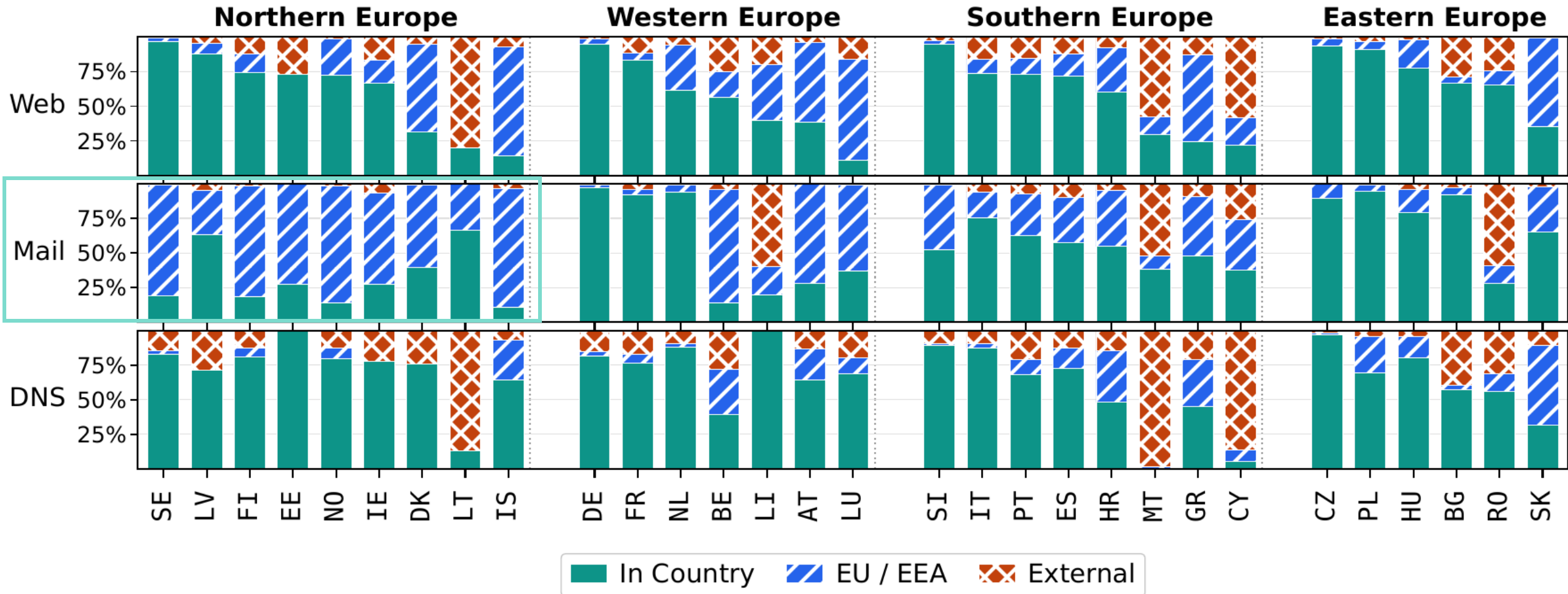
Results 1: Physical Hosting Locations



Results 1: Physical Hosting Locations

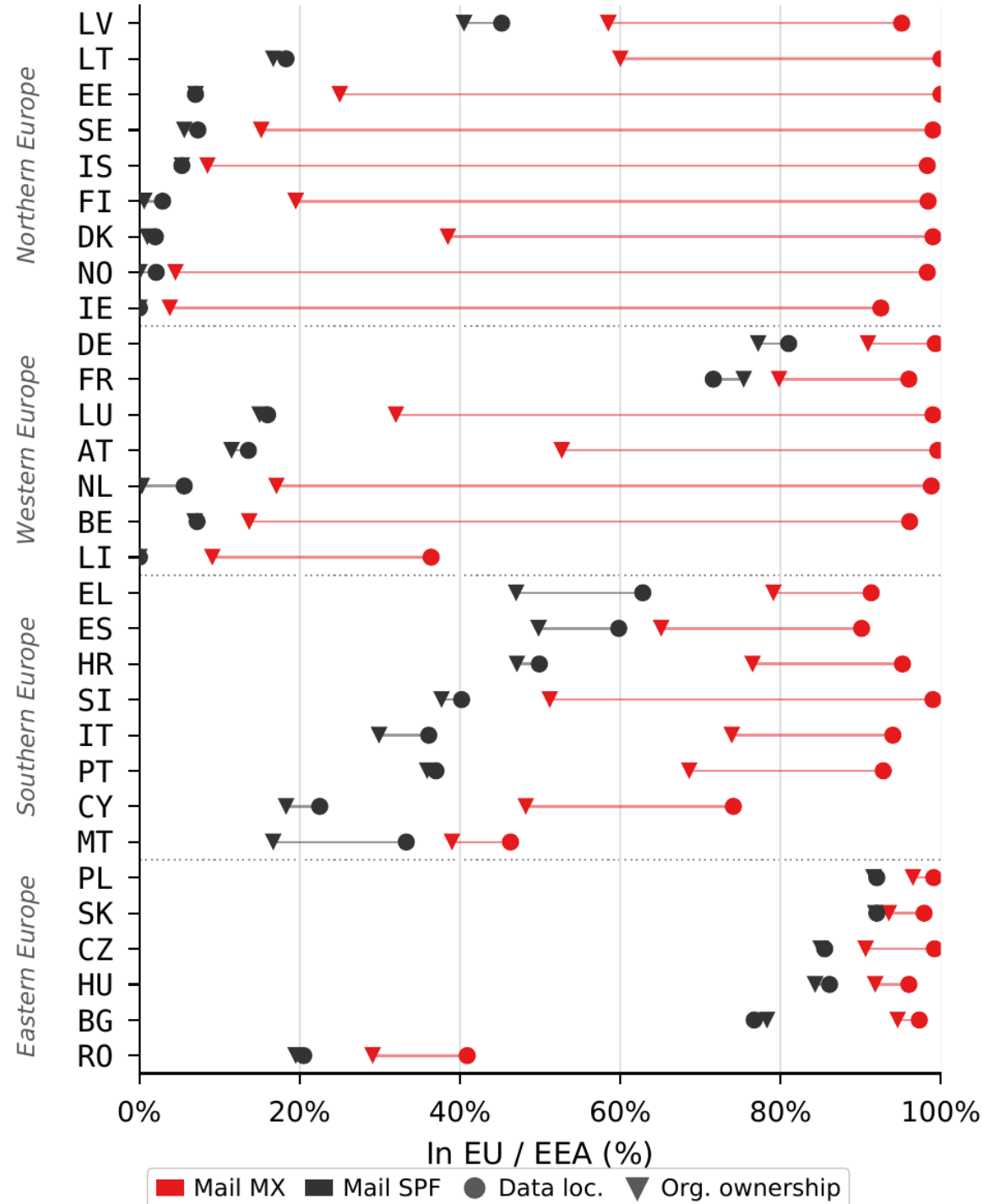


Results: Physical Hosting Locations

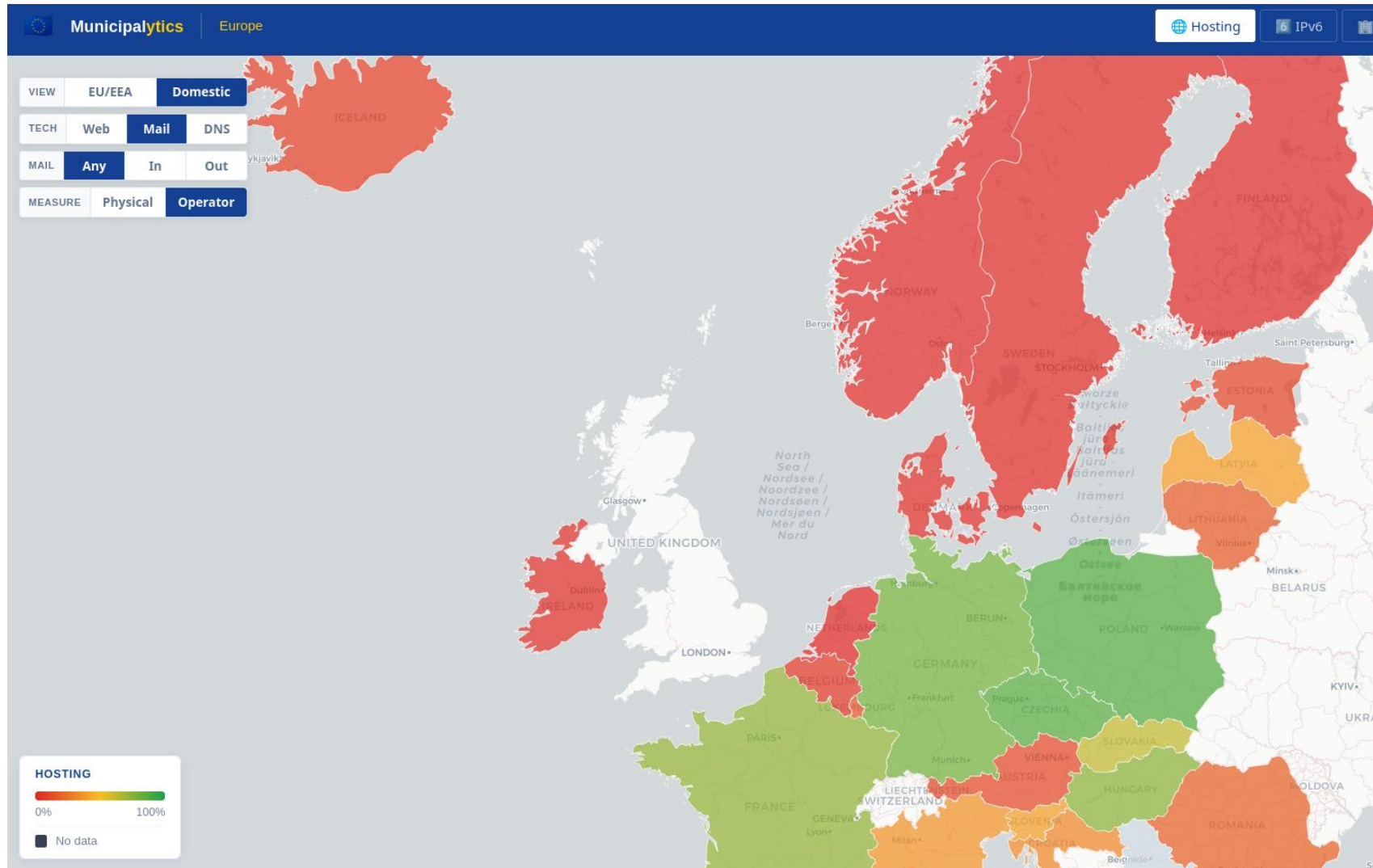


Results: Jurisdictional Dependencies

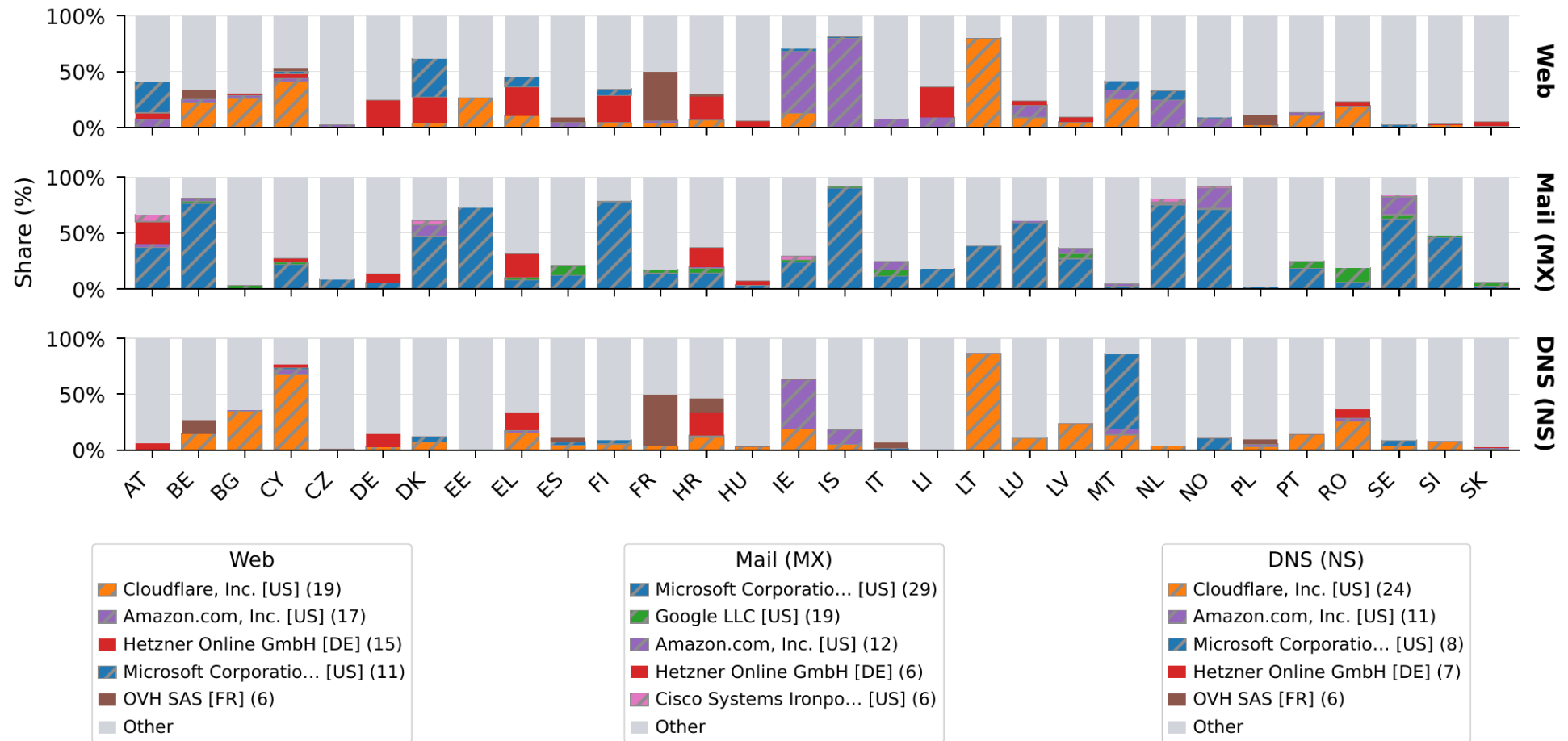
- Wide gap = strong influence by Jurisdiction; Microsoft operates infrastructure in EU, but is still headquartered in the US
- Outbound mail trust is often granted outside EU
- Eastern Europe: In EU inbound and outbound mail (RO exception with strong influence of Yahoo)



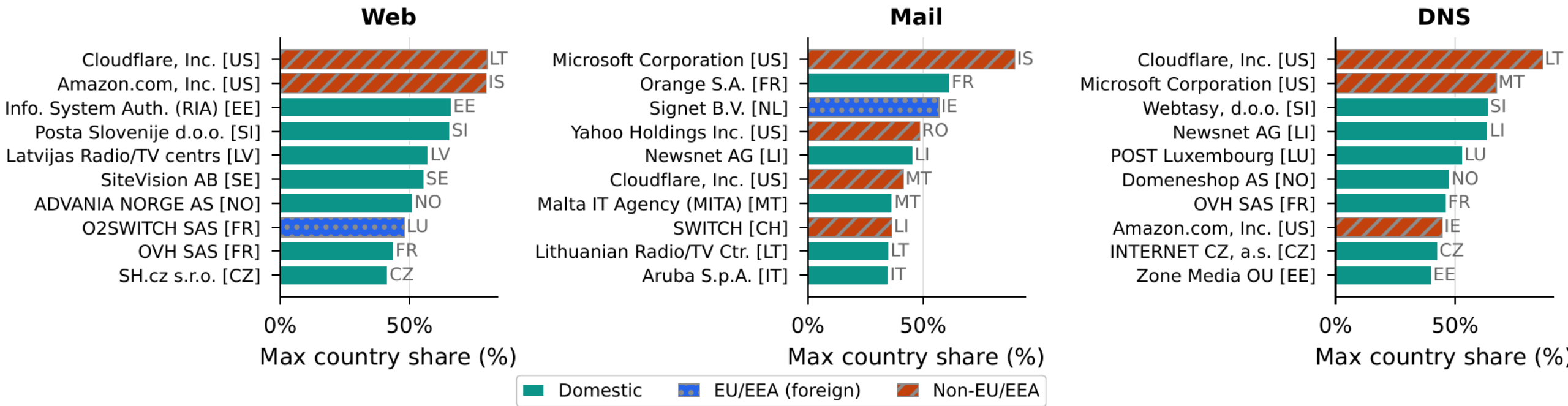
Results: Northern European Mail in the US cloud



Results: Top Providers



Results: Top Providers



Results: Flanders vs. Wallonia



Municipalitytics

Europe >  Belgium



Hosting

6 IPv6



Providers



Has Wikidata



Population

← Back to Europe

NUTS 2

Regions

NUTS 3

Districts

LAU

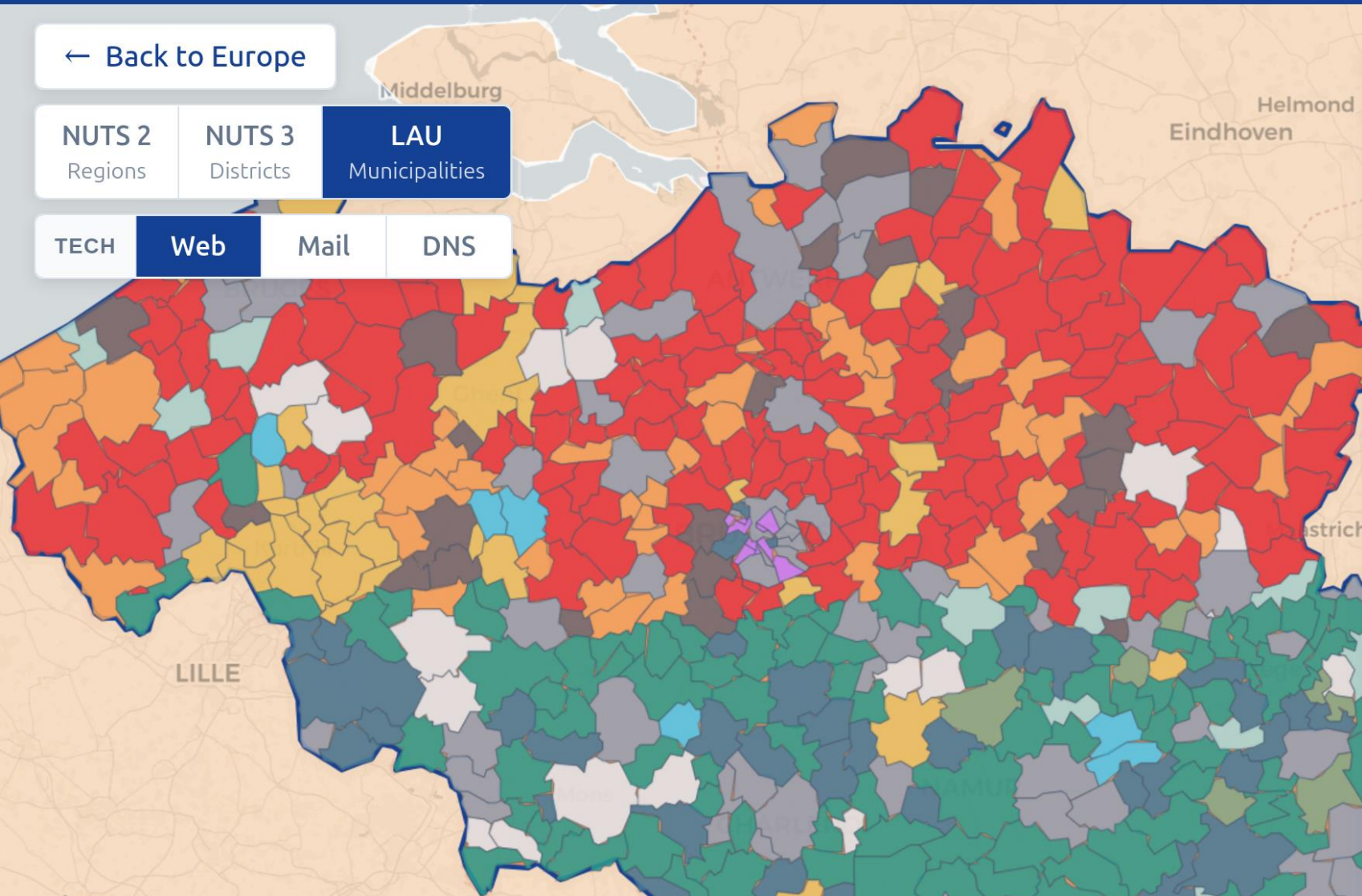
Municipalities

TECH

Web

Mail

DNS



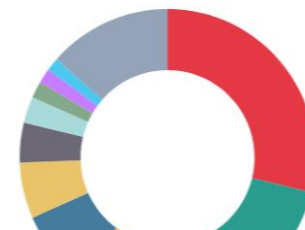
MUNICIPALITY

Attert

Arrondissement of Arlon · 5.797 pop.

Providers	Cloudflare, Inc.
Website	https://www.attert.be/
Web domain	attert.be
Mail domain	attert.be
Area	71.08 km²
DEGURBA	3 · Rural area

PROVIDERS — WEB



Results: Centralization

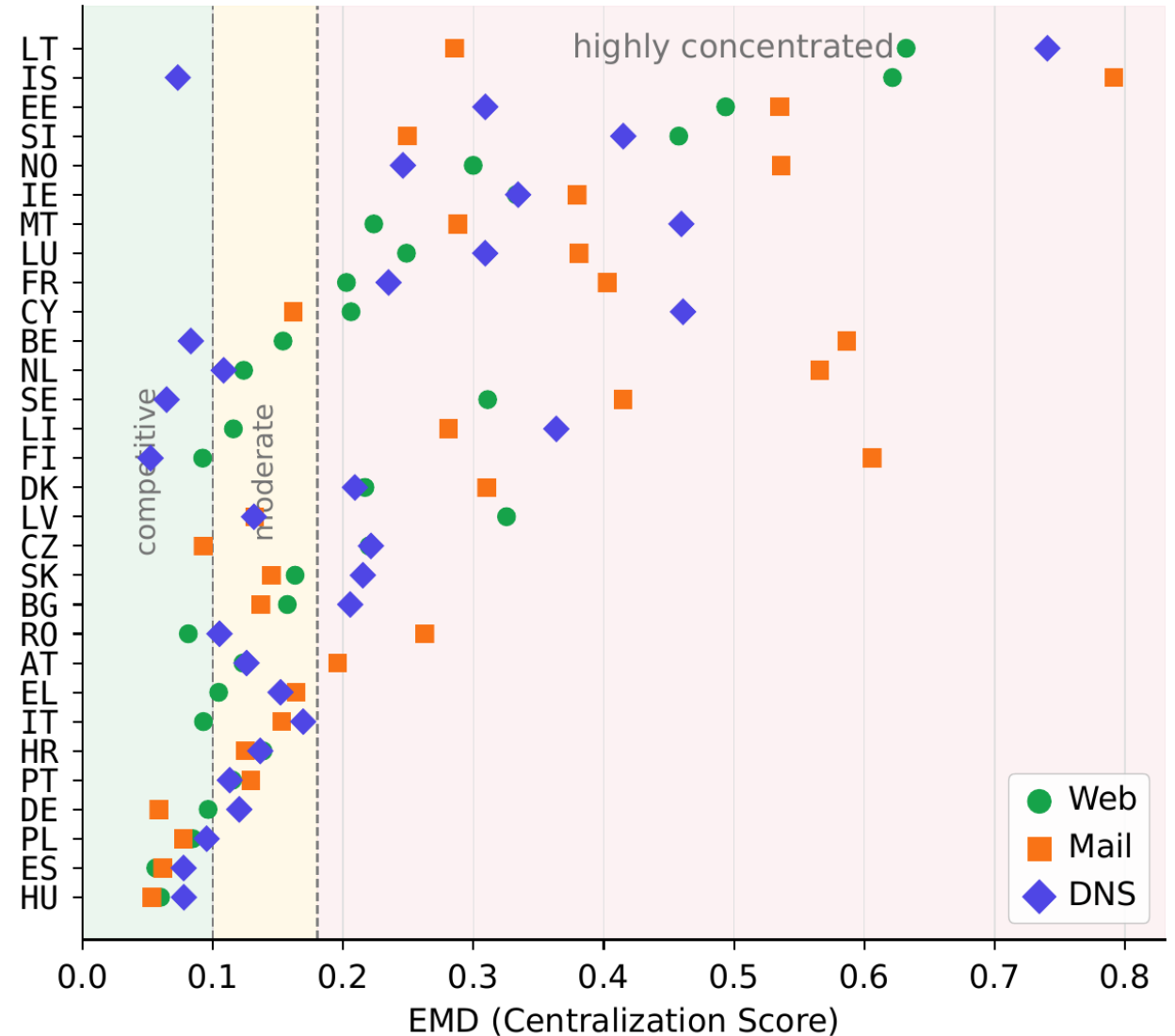
- Earth Mover's Distance (EMD) from IMC'25 *Formalizing Dependence of Web Infrastructure*

$$\mathcal{S} = \sum_{i=0}^n \left(\frac{a_i}{C} \right)^2 - \frac{1}{C}$$

a_i ... # of munici. hosted by provider i
 C ... # of munici. in country

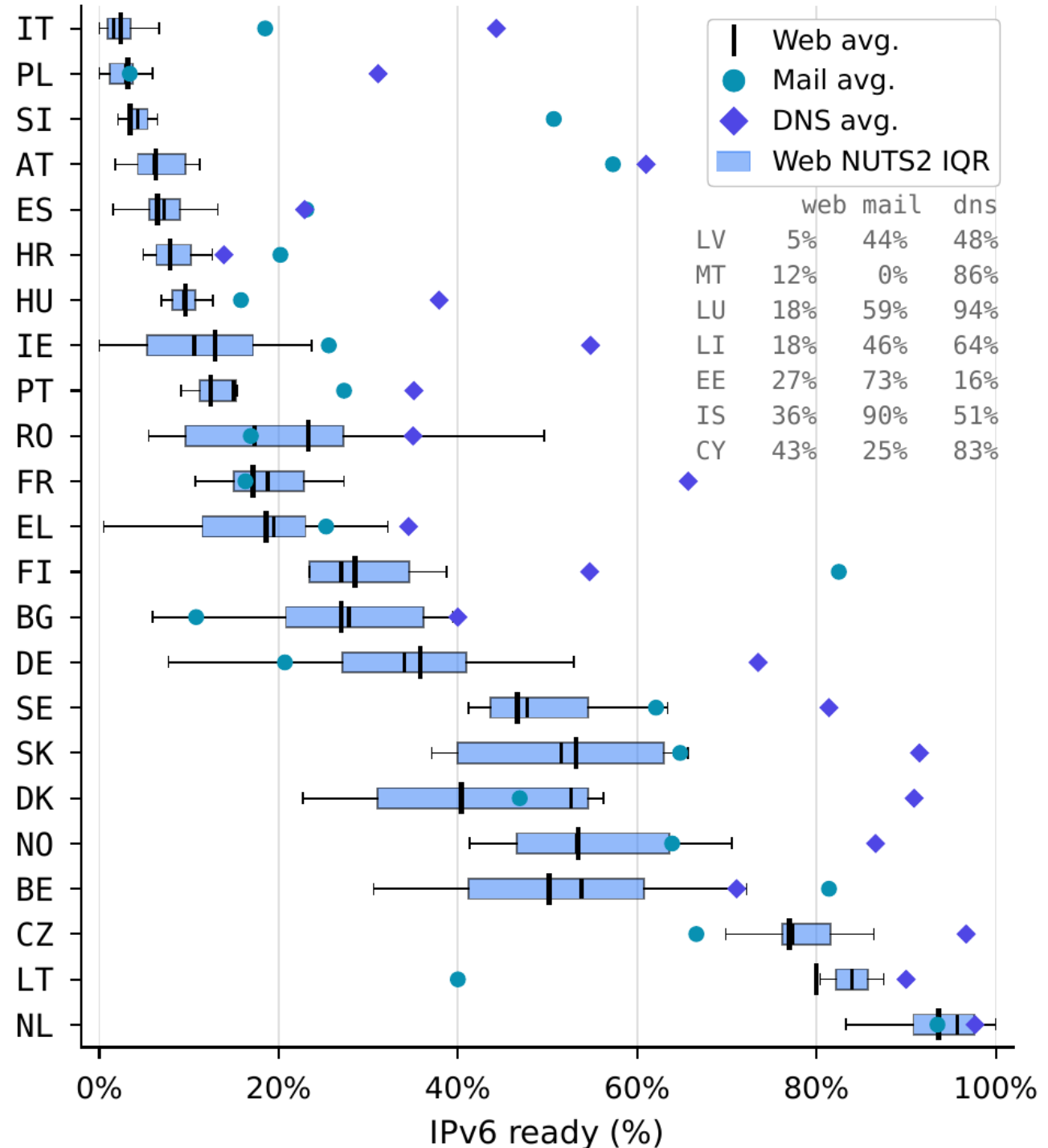
Interpretation (IMC'25):

- *competitive* (<0.10),
- *moderately concentrated* (0.10-0.18)
- *highly concentrated* (>0.18)



Results: IPv6 Reachability

- IPv6 reachability shows largest gap between countries
- Often DNS > Mail > Web
 - o Web is more often hosted domestically
 - o Mail and DNS influenced by centralization
- Effect of national strategies visibile(CZ, NL)



TAKEAWAYS

- 30 Countries - different digital strategies visible
 - US cloud dependency (Northern Europe)
 - Domestic strategies (DE, ES, PL and Eastern europe)
 - IPv6 adoption (NL, CZ)
- Municipal operators rarely cover Web, Mail and DNS
- Digital sovereignty yes, but also market concentration important to resilience

CHECK YOUR COUNTRY



`municipalytics.internetwarte.eu`



interdisciplinary
transformation
university austria